

# Membangun Aplikasi Toko Online Menggunakan Flutter dan Firebase

(Dengan Fitur Upload Gambar Produk pada Dashboard Admin)

Materi ini dirancang untuk level menengah (semester 6) dengan pendekatan **project-based learning**.

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## 1. Learning Outcomes (CPL/CPMK)

Setelah mengikuti materi ini, mahasiswa mampu:

1. Menjelaskan arsitektur aplikasi mobile berbasis **Flutter** dan backend **Firebase**
  2. Mengimplementasikan autentikasi admin dan user menggunakan Firebase Authentication
  3. Mendesain struktur database NoSQL menggunakan Cloud Firestore
  4. Mengupload dan menampilkan gambar produk menggunakan Firebase Storage
  5. Membangun dashboard admin CRUD produk
  6. Mendeploy aplikasi Android yang terintegrasi dengan Firebase
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## 2. Arsitektur Sistem

### 2.1 Arsitektur Teknologi

```
Flutter App
|
|-- Firebase Authentication
|-- Cloud Firestore (Database)
|-- Firebase Storage (Image)
```

### 2.2 Struktur Koleksi Firestore

Collection: `products`

Field:

- `id` (auto)
- `name` (String)
- `description` (String)
- `price` (double)
- `imageUrl` (String)
- `createdAt` (Timestamp)

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## 3. Persiapan Project

### 3.1 Buat Project Flutter

```
flutter create toko_online  
cd toko_online
```

### 3.2 Tambahkan Dependency di pubspec.yaml

```
dependencies:  
  flutter:  
    sdk: flutter  
  firebase_core: ^2.0.0  
  firebase_auth: ^4.0.0  
  cloud_firestore: ^4.0.0  
  firebase_storage: ^11.0.0  
  image_picker: ^1.0.0
```

Kemudian jalankan:

```
flutter pub get
```

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## 4. Konfigurasi Firebase

1. Masuk ke console Firebase
2. Buat project baru
3. Tambahkan aplikasi Android
4. Download google-services.json
5. Letakkan di:

```
android/app/
```

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## 5. Inisialisasi Firebase

### main.dart

```
import 'package:flutter/material.dart';  
import 'package:firebase_core/firebase_core.dart';  
import 'admin_dashboard.dart';  
  
void main() async {  
  WidgetsFlutterBinding.ensureInitialized();  
  await Firebase.initializeApp();  
  runApp(MyApp());  
}
```

```

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: AdminDashboard(),
    );
  }
}

```

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## 6. Dashboard Admin – Upload Produk

### 6.1 UI Form Tambah Produk

#### admin\_dashboard.dart

```

import 'dart:io';
import 'package:flutter/material.dart';
import 'package:image_picker/image_picker.dart';
import 'package:firebase_storage/firebase_storage.dart';
import 'package:cloud_firestore/cloud_firestore.dart';

class AdminDashboard extends StatefulWidget {
  @override
  _AdminDashboardState createState() => _AdminDashboardState();
}

class _AdminDashboardState extends State<AdminDashboard> {

  final TextEditingController nameController = TextEditingController();
  final TextEditingController priceController = TextEditingController();
  final TextEditingController descController = TextEditingController();

  File? _image;
  final picker = ImagePicker();

  Future pickImage() async {
    final pickedFile = await picker.pickImage(source: ImageSource.gallery);
    if (pickedFile != null) {
      setState(() {
        _image = File(pickedFile.path);
      });
    }
  }

  Future uploadProduct() async {
    if (_image == null) return;

    String fileName = DateTime.now().millisecondsSinceEpoch.toString();

    UploadTask uploadTask = FirebaseStorage.instance
      .ref('products/$fileName.jpg')
      .putFile(_image!);

    TaskSnapshot snapshot = await uploadTask;
    String imageUrl = await snapshot.ref.getDownloadURL();
  }
}

```

```

        await FirebaseFirestore.instance.collection('products').add({
          'name': nameController.text,
          'price': double.parse(priceController.text),
          'description': descController.text,
          'imageUrl': imageUrl,
          'createdAt': Timestamp.now(),
        });

        ScaffoldMessenger.of(context).showSnackBar(
          SnackBar(content: Text("Produk berhasil ditambahkan"))
        );
      }

    @override
    Widget build(BuildContext context) {
      return Scaffold(
        appBar: AppBar(title: Text("Dashboard Admin")),
        body: SingleChildScrollView(
          padding: EdgeInsets.all(16),
          child: Column(
            children: [
              TextField(
                controller: nameController,
                decoration: InputDecoration(labelText: "Nama Produk"),
              ),
              TextField(
                controller: priceController,
                decoration: InputDecoration(labelText: "Harga"),
                keyboardType: TextInputType.number,
              ),
              TextField(
                controller: descController,
                decoration: InputDecoration(labelText: "Deskripsi"),
              ),
              SizedBox(height: 10),
              _image == null
                ? Text("Belum ada gambar")
                : Image.file(_image!, height: 150),
              ElevatedButton(
                onPressed: pickImage,
                child: Text("Pilih Gambar"),
              ),
              SizedBox(height: 10),
              ElevatedButton(
                onPressed: uploadProduct,
                child: Text("Upload Produk"),
              ),
            ],
          ),
        ),
      );
    }
  }
}

```

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## 7. Menampilkan Produk di Halaman User

## product\_list.dart

```
import 'package:flutter/material.dart';
import 'package:cloud_firestore/cloud_firestore.dart';

class ProductList extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text("Daftar Produk")),
      body: StreamBuilder(
        stream: FirebaseFirestore.instance
          .collection('products')
          .orderBy('createdAt', descending: true)
          .snapshots(),
        builder: (context, AsyncSnapshot<QuerySnapshot> snapshot) {
          if (!snapshot.hasData) return CircularProgressIndicator();

          return ListView(
            children: snapshot.data!.docs.map((doc) {
              return Card(
                child: ListTile(
                  leading: Image.network(doc['imageUrl']),
                  title: Text(doc['name']),
                  subtitle: Text("Rp ${doc['price']}"),
                ),
              );
            }).toList(),
          );
        },
      ),
    );
  }
}
```

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## 8. Praktikum Project-Based Learning

### Project Tugas Akhir Mahasiswa:

Mahasiswa wajib menambahkan:

1. Fitur Edit Produk
  2. Fitur Hapus Produk
  3. Autentikasi Admin Login
  4. Fitur Keranjang Belanja
  5. Checkout sederhana
  6. Validasi input form
  7. State management (Provider / Riverpod)
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## 9. Rubrik Penilaian

| Aspek              | Bobot |
|--------------------|-------|
| UI/UX              | 20%   |
| CRUD Produk        | 25%   |
| Upload Gambar      | 20%   |
| Integrasi Firebase | 20%   |
| Presentasi & Demo  | 15%   |

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## 10. Pengembangan Lanjutan (Research-Oriented)

Untuk level riset atau publikasi:

- Implementasi Clean Architecture
- Integrasi Payment Gateway
- Implementasi caching & offline mode
- Analisis performa Firebase Storage
- Perbandingan Firestore vs Realtime DB